# Matrices: Questions and Full Solutions

## Question 01

**English**

Let $P$ and $Q$ be the $2\times 2$ matrices of reflection of the $xy$-plane in the lines through the origin making angles $\frac{\pi}{8}$ and $\frac{\pi}{3}$, respectively, with the positive $x$-axis. Let $I$ denote the $2\times 2$ identity matrix. Then the number of positive integers $n \le 60$ for which $(QP)^n + (PQ)^n = I$ is

(A) 2
(B) 3
(C) 5
(D) 10

Only one option is correct.

**हिन्दी**

माना कि $P$ तथा $Q$, $xy$-तल के उन $2 \times 2$ आव्यूहों को निरूपित करते हैं जो क्रमशः मूलबिन्दु से होकर जाने वाली तथा धनात्मक $x$-अक्ष से $\frac{\pi}{8}$ तथा $\frac{\pi}{3}$ कोण बनाने वाली रेखाओं में परावर्तन दर्शाते हैं। माना $I$, $2 \times 2$ तत्समक आव्यूह है। तब उन धनात्मक पूर्णांकों $n \le 60$ की संख्या, जिनके लिए $(QP)^n + (PQ)^n = I$ है, होगी

(A) 2
(B) 3
(C) 5
(D) 10

केवल एक विकल्प सही है।

**Solution**

The reflection of the plane in the line through the origin at angle $\alpha$ with the $x$-axis has matrix $F_\alpha = \begin{pmatrix} \cos 2\alpha & \sin 2\alpha \\ \sin 2\alpha & -\cos 2\alpha \end{pmatrix}$ (it fixes $(\cos\alpha, \sin\alpha)$ and reverses $(-\sin\alpha, \cos\alpha)$).

So $P = F_{\pi/8}$ and $Q = F_{\pi/3}$.

Multiplying, the $(1,1)$ entry of $F_\beta F_\alpha$ is $\cos 2\beta \cos 2\alpha + \sin 2\beta \sin 2\alpha = \cos 2(\beta - \alpha)$ and the $(2,1)$ entry is $\sin 2\beta \cos 2\alpha - \cos 2\beta \sin 2\alpha = \sin 2(\beta - \alpha)$, by the compound-angle formulas.

Similarly the other entries give $F_\beta F_\alpha = \begin{pmatrix} \cos 2(\beta-\alpha) & -\sin 2(\beta-\alpha) \\ \sin 2(\beta-\alpha) & \cos 2(\beta-\alpha) \end{pmatrix} = R_{2(\beta-\alpha)}$, the rotation through $2(\beta - \alpha)$.

Hence $QP = R_\theta$ with $\theta = 2\left(\frac{\pi}{3} - \frac{\pi}{8}\right) = \frac{5\pi}{12}$, and $PQ = R_{2(\pi/8 - \pi/3)} = R_{-\theta}$.

The matrices $R_\theta$ multiply exactly as the complex numbers $\cos\theta + i\sin\theta$ do: $R_\theta R_\varphi = R_{\theta+\varphi}$. By De Moivre's theorem $(\cos\theta + i\sin\theta)^n = \cos n\theta + i \sin n\theta$, so $(QP)^n = R_{n\theta}$ and $(PQ)^n = R_{-n\theta}$.

Adding, the sine terms cancel: $(QP)^n + (PQ)^n = \begin{pmatrix} 2\cos n\theta & 0 \\ 0 & 2\cos n\theta \end{pmatrix} = 2\cos(n\theta)\, I$.

So the condition is $\cos\frac{5n\pi}{12} = \frac{1}{2}$, i.e. $\frac{5n\pi}{12} = 2k\pi \pm \frac{\pi}{3}$ for an integer $k$.

Multiplying by $\frac{12}{\pi}$: $5n = 24k \pm 4$, i.e. $5n \equiv \pm 4 \pmod{24}$.

Since $5 \cdot 5 = 25 \equiv 1 \pmod{24}$, multiplying by 5 gives $n \equiv \pm 20 \pmod{24}$, i.e. $n \equiv 4$ or $n \equiv 20 \pmod{24}$.

For $1 \le n \le 60$: $n \equiv 4$ gives $4, 28, 52$ and $n \equiv 20$ gives $20, 44$. Hence there are 5 values.

(Counting $\cos\dfrac{5n\pi}{12} = \pm\dfrac{1}{2}$ instead would give 10, and keeping only one residue class gives 3 or 2.)

**Correct Option: (C)**

## Question 02

**English**

Let $A$ be a $3\times 3$ real matrix such that $\mathrm{tr}(A) = 6$, $\mathrm{tr}\left(A^2\right) = 14$ and $\det(A) = 6$, where $\mathrm{tr}(X)$ denotes the trace of a square matrix $X$. If $\mathrm{adj}(A) = A^2 + \beta A + \gamma I$ for some real numbers $\beta$ and $\gamma$, where $I$ is the $3 \times 3$ identity matrix, then $(\beta, \gamma)$ is equal to

(A) $(-6, 11)$
(B) $(6, 11)$
(C) $(-6, -11)$
(D) $(-6, 14)$

Only one option is correct.

**हिन्दी**

माना कि $A$ एक $3 \times 3$ वास्तविक आव्यूह है जिसके लिए $\mathrm{tr}(A) = 6$, $\mathrm{tr}\left(A^2\right) = 14$ तथा $\det(A) = 6$ है, जहाँ $\mathrm{tr}(X)$ वर्ग आव्यूह $X$ का अनुरेख (trace) है। यदि किन्हीं वास्तविक संख्याओं $\beta$ तथा $\gamma$ के लिए $\mathrm{adj}(A) = A^2 + \beta A + \gamma I$ है, जहाँ $I$, $3 \times 3$ तत्समक आव्यूह है, तो $(\beta, \gamma)$ बराबर है

(A) $(-6, 11)$
(B) $(6, 11)$
(C) $(-6, -11)$
(D) $(-6, 14)$

केवल एक विकल्प सही है।

**Solution**

Let $\lambda_1, \lambda_2, \lambda_3$ be the eigenvalues of $A$ (roots of $\det(A - \lambda I) = 0$). Then $\mathrm{tr}(A) = \sum \lambda_i = 6$, $\mathrm{tr}(A^2) = \sum \lambda_i^2 = 14$ and $\det(A) = \lambda_1\lambda_2\lambda_3 = 6$.

From $(\sum \lambda_i)^2 = \sum \lambda_i^2 + 2\sum \lambda_i\lambda_j$: $\sum \lambda_i\lambda_j = \dfrac{36 - 14}{2} = 11$.

So the characteristic equation is $\lambda^3 - 6\lambda^2 + 11\lambda - 6 = 0$, i.e. $(\lambda-1)(\lambda-2)(\lambda-3) = 0$; the eigenvalues are $1, 2, 3$.

By the Cayley-Hamilton theorem, $A^3 - 6A^2 + 11A - 6I = O$.

Rewrite it as $A\left(A^2 - 6A + 11I\right) = 6I$.

Since $\det(A) = 6 \neq 0$, $A$ is invertible and $A^{-1} = \dfrac{1}{6}\left(A^2 - 6A + 11I\right)$.

Also $A\,\mathrm{adj}(A) = \det(A)\, I = 6I$, so $\mathrm{adj}(A) = 6A^{-1} = A^2 - 6A + 11I$.

Uniqueness: if $A^2 + \beta A + \gamma I = A^2 + \beta' A + \gamma' I$ with $(\beta, \gamma) \neq (\beta', \gamma')$, then $A$ would satisfy a non-zero polynomial of degree at most 1, so $A$ would have at most one eigenvalue; but $A$ has three distinct eigenvalues. Hence $\beta = -6$, $\gamma = 11$.

(Taking $\sum \lambda_i\lambda_j = 14$ or a wrong sign gives the other options.)

**Correct Option: (A)**

## Question 03

**English**

Let $Q = \frac{1}{3}\begin{pmatrix} 1 & 2 & 2 \\ 2 & 1 & -2 \\ 2 & -2 & 1 \end{pmatrix}$ and $X = \begin{pmatrix} 1 & 2 & 0 \\ 0 & 1 & -1 \\ 2 & 0 & 1 \end{pmatrix}$, and let $M = QX^TQ^T$, where $Y^T$ denotes the transpose of a matrix $Y$. Then the sum of the squares of all the entries of $M$ is

(A) 9
(B) 12
(C) 16
(D) 36

Only one option is correct.

**हिन्दी**

माना कि $Q = \frac{1}{3}\begin{pmatrix} 1 & 2 & 2 \\ 2 & 1 & -2 \\ 2 & -2 & 1 \end{pmatrix}$ तथा $X = \begin{pmatrix} 1 & 2 & 0 \\ 0 & 1 & -1 \\ 2 & 0 & 1 \end{pmatrix}$ है, और $M = QX^TQ^T$ है, जहाँ $Y^T$ आव्यूह $Y$ का परिवर्त (transpose) है। तब $M$ की सभी प्रविष्टियों के वर्गों का योग है

(A) 9
(B) 12
(C) 16
(D) 36

केवल एक विकल्प सही है।

**Solution**

For any square matrix $Y = [y_{ij}]$, the $(j, j)$ entry of $Y^TY$ is $\sum_i y_{ij}^2$, so $\operatorname{tr}\left(Y^TY\right) = \sum_{i,j} y_{ij}^2$, the sum of the squares of all entries.

The rows of $Q$ are $\frac{1}{3}(1, 2, 2)$, $\frac{1}{3}(2, 1, -2)$, $\frac{1}{3}(2, -2, 1)$. Each has length $\frac{1}{3}\sqrt{1+4+4} = 1$, and the dot products are $\frac{1}{9}(2+2-4) = 0$, $\frac{1}{9}(2-4+2) = 0$, $\frac{1}{9}(4-2-2) = 0$. Hence $QQ^T = I$, so $Q^T = Q^{-1}$ and $Q^TQ = I$.

By $(ABC)^T = C^TB^TA^T$: $M^T = \left(QX^TQ^T\right)^T = \left(Q^T\right)^T\left(X^T\right)^TQ^T = QXQ^T$.

Then $M^TM = QXQ^T \cdot QX^TQ^T = QX\left(Q^TQ\right)X^TQ^T = Q\left(XX^T\right)Q^T$.

Using $\operatorname{tr}(UV) = \operatorname{tr}(VU)$ with $U = Q$, $V = XX^TQ^T$: $\operatorname{tr}\left(M^TM\right) = \operatorname{tr}\left(XX^TQ^TQ\right) = \operatorname{tr}\left(XX^T\right)$.

$\operatorname{tr}\left(XX^T\right)$ is the sum of the squares of the entries of $X$: $1^2+2^2+0+0+1^2+(-1)^2+2^2+0+1^2 = 12$.

So the required sum is 12 (a direct multiplication of the three matrices gives the same value).

**Correct Option: (B)**

## Question 04

**English**

For a real number $\lambda$, consider the system of linear equations

$$x + 2y + 3z = \lambda x, \qquad 2x + 4y + 6z = \lambda y, \qquad 3x + 6y + 9z = \lambda z$$

in the unknowns $x, y, z$. Then the set of all values of $\lambda$ for which the system has a non-trivial solution is

(A) $\{14\}$

(B) $\{0\}$
(C) $\{0, -14\}$
(D) $\{0, 14\}$

Only one option is correct.

**हिन्दी**

वास्तविक संख्या $\lambda$ के लिए अज्ञात $x, y, z$ में रैखिक समीकरण निकाय

$$x + 2y + 3z = \lambda x, \qquad 2x + 4y + 6z = \lambda y, \qquad 3x + 6y + 9z = \lambda z$$

पर विचार कीजिए। तब $\lambda$ के उन सभी मानों का समुच्चय, जिनके लिए निकाय का एक अतुच्छ (non-trivial) हल है, होगा

(A) $\{14\}$
(B) $\{0\}$
(C) $\{0, -14\}$
(D) $\{0, 14\}$

केवल एक विकल्प सही है।

**Solution**

Moving the right sides to the left, the system is $(1-\lambda)x + 2y + 3z = 0$, $2x + (4-\lambda)y + 6z = 0$, $3x + 6y + (9-\lambda)z = 0$, a homogeneous system $(C - \lambda I)X = O$ with $C = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 3 & 6 & 9 \end{pmatrix}$.

A homogeneous square system has a non-trivial solution exactly when its coefficient determinant is 0.

Expanding, $\det(C - \lambda I) = -\lambda^3 + (\operatorname{tr} C)\lambda^2 - m\lambda + \det C$, where $m$ is the sum of the three $2 \times 2$ principal minors of $C$.

Each row of $C$ is a multiple of $(1, 2, 3)$, so every $2 \times 2$ minor is 0 (for example $1 \cdot 4 - 2 \cdot 2 = 0$, $1 \cdot 9 - 3 \cdot 3 = 0$, $4 \cdot 9 - 6 \cdot 6 = 0$) and $\det C = 0$.

With $\operatorname{tr} C = 1 + 4 + 9 = 14$: $\det(C - \lambda I) = -\lambda^3 + 14\lambda^2 = \lambda^2(14 - \lambda)$.

So the determinant vanishes only for $\lambda = 0$ and $\lambda = 14$.

Check $\lambda = 0$: every equation reduces to $x + 2y + 3z = 0$, which has non-trivial solutions such as $(2, -1, 0)$.

Check $\lambda = 14$: $(x, y, z) = (1, 2, 3)$ satisfies all three equations ($1 + 4 + 9 = 14$, $2 + 8 + 18 = 28$, $3 + 12 + 27 = 42$).

Hence the set is $\{0, 14\}$.

**Correct Option: (D)**

## Question 05

**English**

Let $a, b, c$ be the roots of the equation $t^3 - 3t^2 + 1 = 0$, and let $A = \begin{pmatrix} a & b & c \\ b & c & a \\ c & a & b \end{pmatrix}$. Then $\det(\operatorname{adj} A) + \det A$ is equal to

(A) 702
(B) 756
(C) $-702$

(D) 0

Only one option is correct.

**हिन्दी**

माना कि $a, b, c$ समीकरण $t^3 - 3t^2 + 1 = 0$ के मूल हैं, तथा $A = \begin{pmatrix} a & b & c \\ b & c & a \\ c & a & b \end{pmatrix}$ है। तब $\det(\operatorname{adj} A) + \det A$ बराबर है

(A) 702
(B) 756
(C) $-702$
(D) 0

केवल एक विकल्प सही है।

**Solution**

Apply $C_1 \to C_1 + C_2 + C_3$; every entry of the new first column is $a + b + c$, so $\det A = (a + b + c)\begin{vmatrix} 1 & b & c \\ 1 & c & a \\ 1 & a & b \end{vmatrix}$.

Apply $R_2 \to R_2 - R_1$ and $R_3 \to R_3 - R_1$: the determinant becomes $\begin{vmatrix} 1 & b & c \\ 0 & c-b & a-c \\ 0 & a-b & b-c \end{vmatrix} = (c-b)(b-c) - (a-c)(a-b)$.

Expanding: $-(b-c)^2 - (a^2 - ab - ac + bc) = -\left(a^2 + b^2 + c^2 - ab - bc - ca\right)$.

So $\det A = -(a+b+c)\left(a^2 + b^2 + c^2 - ab - bc - ca\right) = -\left(a^3 + b^3 + c^3 - 3abc\right)$; this is symmetric in $a, b, c$, so the order of the roots does not matter.

By Vieta's relations for $t^3 - 3t^2 + 0 \cdot t + 1 = 0$: $a + b + c = 3$, $ab + bc + ca = 0$, $abc = -1$.

Then $a^2 + b^2 + c^2 = (a+b+c)^2 - 2(ab + bc + ca) = 9$, and $\det A = -3(9 - 0) = -27$.

For an $n \times n$ matrix, $A \operatorname{adj} A = (\det A) I$ gives $\det A \cdot \det(\operatorname{adj} A) = (\det A)^n$, so $\det(\operatorname{adj} A) = (\det A)^{n-1}$ when $\det A \neq 0$.

With $n = 3$: $\det(\operatorname{adj} A) = (-27)^2 = 729$.

Hence $\det(\operatorname{adj} A) + \det A = 729 - 27 = 702$. (Taking $\det A = +27$ would give 756.)

**Correct Option: (A)**

## Question 06

**English**

Let $\alpha < \beta < \gamma$ be the roots of the equation $x^3 - 7x + 6 = 0$, and let $A = \begin{pmatrix} 1 & \alpha & \alpha^2 \\ 1 & \beta & \beta^2 \\ 1 & \gamma & \gamma^2 \end{pmatrix}$. Then the sum of all the entries of $\text{adj}(A)$ is

(A) 1
(B) 20
(C) $-20$
(D) 60

Only one option is correct.

**हिन्दी**

माना कि $\alpha < \beta < \gamma$ समीकरण $x^3 - 7x + 6 = 0$ के मूल हैं, तथा $A = \begin{pmatrix} 1 & \alpha & \alpha^2 \\ 1 & \beta & \beta^2 \\ 1 & \gamma & \gamma^2 \end{pmatrix}$ है। तब $\text{adj}(A)$ की सभी प्रविष्टियों का योग है

(A) 1
(B) 20
(C) $-20$
(D) 60

केवल एक विकल्प सही है।

**Solution**

$x = 1$ is a root, and $x^3 - 7x + 6 = (x-1)(x^2 + x - 6) = (x-1)(x-2)(x+3)$. With $\alpha < \beta < \gamma$: $\alpha = -3$, $\beta = 1$, $\gamma = 2$.

Apply $R_2 \to R_2 - R_1$, $R_3 \to R_3 - R_1$: the rows become $(0, \beta - \alpha, \beta^2 - \alpha^2)$ and $(0, \gamma - \alpha, \gamma^2 - \alpha^2)$. Taking $(\beta - \alpha)$ and $(\gamma - \alpha)$ out of these rows leaves $\begin{vmatrix} 1 & \beta + \alpha \\ 1 & \gamma + \alpha \end{vmatrix} = \gamma - \beta$.

So $\det A = (\beta - \alpha)(\gamma - \alpha)(\gamma - \beta) = (1+3)(2+3)(2-1) = 20 \neq 0$, and $A$ is invertible.

The first column of $A$ is $(1, 1, 1)^T$, i.e. $A\, e_1 = (1, 1, 1)^T$ with $e_1 = (1, 0, 0)^T$.

Multiplying by $A^{-1}$: $A^{-1}(1, 1, 1)^T = e_1$. The left side is the column of row sums of $A^{-1}$, so the row sums of $A^{-1}$ are $1, 0, 0$.

Hence the sum of all entries of $A^{-1}$ is $1 + 0 + 0 = 1$.

Since $\text{adj}(A) = \det(A)\, A^{-1} = 20A^{-1}$, the sum of all entries of $\text{adj}(A)$ is 20.

(Direct check: the cofactor matrix has entries $30, -5, -5$; $-12, 8, 4$; $2, -3, 1$, whose sum is 20.)

**Correct Option: (B)**

## Question 07

**English**

Let $N = \begin{pmatrix} 4 & 6 & 1 \\ 0 & 1 & 12 \\ 0 & 0 & 9 \end{pmatrix}$, and let $S$ denote the set of all $3 \times 3$ upper triangular matrices $X$ with real entries such that $X^2 = N$. Then the least value of the sum of all the entries of $X$, as $X$ varies over $S$, is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

**हिन्दी**

माना कि $N = \begin{pmatrix} 4 & 6 & 1 \\ 0 & 1 & 12 \\ 0 & 0 & 9 \end{pmatrix}$ है, तथा $S$ वास्तविक प्रविष्टियों वाले उन सभी $3 \times 3$ उपरि त्रिभुजीय आव्यूहों $X$ का समुच्चय है जिनके लिए $X^2 = N$ है। तब $X$ के $S$ में परिवर्तित होने पर $X$ की सभी प्रविष्टियों के योग का न्यूनतम मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

**Solution**

Let $X = \begin{pmatrix} a & p & q \\ 0 & b & r \\ 0 & 0 & c \end{pmatrix}$. The product of two upper triangular matrices is upper triangular, and row-by-column multiplication gives

$$X^2 = \begin{pmatrix} a^2 & p(a+b) & aq+pr+qc \\ 0 & b^2 & r(b+c) \\ 0 & 0 & c^2 \end{pmatrix}.$$

Comparing with $N$: $a^2 = 4$, $b^2 = 1$, $c^2 = 9$, so $a = \pm 2$, $b = \pm 1$, $c = \pm 3$.

Since $|a|, |b|, |c|$ are distinct, $a+b$, $b+c$, $a+c$ are never 0. Hence $p(a+b) = 6$, $r(b+c) = 12$ and $q(a+c) + pr = 1$ give unique $p = \dfrac{6}{a+b}$, $r = \dfrac{12}{b+c}$, $q = \dfrac{1-pr}{a+c}$ for each sign choice. So $S$ has exactly 8 elements.

$(2, 1, 3)$: $p = 2,\ r = 3,\ q = \dfrac{1-6}{5} = -1$; sum $= 2+1+3+2+3-1 = 10$.

$(2, 1, -3)$: $p = 2,\ r = -6,\ q = \dfrac{1+12}{-1} = -13$; sum $= 0+2-6-13 = -17$.

$(2, -1, 3)$: $p = 6,\ r = 6,\ q = \dfrac{1-36}{5} = -7$; sum $= 4+6+6-7 = 9$.

$(2, -1, -3)$: $p = 6,\ r = -3,\ q = \dfrac{1+18}{-1} = -19$; sum $= -2+6-3-19 = -18$.

$(-2, 1, 3)$: $p = -6,\ r = 3,\ q = \dfrac{1+18}{1} = 19$; sum $= 2-6+3+19 = 18$.

$(-2, 1, -3)$: $p = -6,\ r = -6,\ q = \dfrac{1-36}{-5} = 7$; sum $= -4-6-6+7 = -9$.

$(-2, -1, 3)$: $p = -2,\ r = 6,\ q = \dfrac{1+12}{1} = 13$; sum $= 0-2+6+13 = 17$.

$(-2, -1, -3)$: $p = -2,\ r = -3,\ q = \dfrac{1-6}{-5} = 1$; sum $= -6-2-3+1 = -10$.

Each of these satisfies $X^2 = N$ (for instance $(2, -1, -3)$: $p(a+b) = 6$, $r(b+c) = 12$, $q(a+c)+pr = 19-18 = 1$).

The least sum is $-18$, attained at $X = \begin{pmatrix} 2 & 6 & -19 \\ 0 & -1 & -3 \\ 0 & 0 & -3 \end{pmatrix}$; note that the choice with all diagonal entries negative gives only $-10$.

**Final Answer: -18**

## Question 08

**English**

Let $S$ denote the set of all $3 \times 3$ matrices $A$ with integer entries such that $A$ is symmetric and $AA^T = I$, where $I$ is the $3 \times 3$ identity matrix. Then the number of elements of $S$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

**हिन्दी**

माना कि $S$ पूर्णांक प्रविष्टियों वाले उन सभी $3 \times 3$ आव्यूहों $A$ का समुच्चय है जिनके लिए $A$ सममित है तथा $AA^T = I$ है, जहाँ $I$, $3 \times 3$ तत्समक आव्यूह है। तब $S$ के अवयवों की संख्या _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

**Solution**

From $AA^T = I$, the $(i, i)$ entry gives $a_{i1}^2 + a_{i2}^2 + a_{i3}^2 = 1$. With integer entries, each row has exactly one entry equal to $\pm 1$ and the other two equal to 0.

The $(i, k)$ entry for $i \neq k$ gives that rows $i$ and $k$ are perpendicular; if their non-zero entries were in the same column the dot product would be $\pm 1$. So the non-zero entries lie in different columns.

Hence $A$ is a signed permutation matrix: $a_{i\sigma(i)} = \varepsilon_i \in \{1, -1\}$ for a permutation $\sigma$ of $\{1, 2, 3\}$, all other entries 0. Every such matrix satisfies $AA^T = I$.

Symmetry $a_{ij} = a_{ji}$: the non-zero pattern must be symmetric, so $\sigma(\sigma(i)) = i$ for all $i$, i.e. $\sigma$ is the identity or a transposition; a 3-cycle is excluded.

Identity permutation: $A = \text{diag}(\varepsilon_1, \varepsilon_2, \varepsilon_3)$, always symmetric: $2^3 = 8$ matrices.

Transposition swapping $i, j$ and fixing $k$ (three choices of the pair): symmetry forces $a_{ij} = a_{ji} = \varepsilon \in \{\pm 1\}$, and $a_{kk} = \pm 1$, giving $2 \times 2 = 4$ matrices per pair, so $3 \times 4 = 12$.

For example $\begin{pmatrix} 0 & -1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}$ is in $S$, while $\begin{pmatrix} 0 & 1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}$ is orthogonal but not symmetric.

Total: $8 + 12 = 20$.

**Final Answer: 20**

## Question 09

**English**

Let $u = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$ and $v = \begin{pmatrix} 2 \\ -1 \\ 2 \end{pmatrix}$, and let $M = I + uv^T$, where $I$ is the $3 \times 3$ identity matrix and $v^T$ is the transpose of $v$. If $\left(M^T\right)^{-1} = I + \lambda\, vu^T$ for a real number $\lambda$, then the value of $\lambda$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

**हिन्दी**

माना कि $u = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$ तथा $v = \begin{pmatrix} 2 \\ -1 \\ 2 \end{pmatrix}$ है, और $M = I + uv^T$ है, जहाँ $I$, $3 \times 3$ तत्समक आव्यूह है तथा $v^T$, $v$ का परिवर्त है। यदि किसी वास्तविक संख्या $\lambda$ के लिए $\left(M^T\right)^{-1} = I + \lambda\, vu^T$ है, तो $\lambda$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

**Solution**

By the transpose rules $(X+Y)^T = X^T + Y^T$ and $(XY)^T = Y^T X^T$: $M^T = I + \left(v^T\right)^T u^T = I + vu^T$.

Here $vu^T$ is a $3 \times 3$ matrix and $u^T v$ is a $1 \times 1$ matrix, i.e. the number $1 \cdot 2 + 1 \cdot (-1) + 1 \cdot 2 = 3$.

By associativity, $\left(vu^T\right)\left(vu^T\right) = v\left(u^T v\right)u^T = 3\, vu^T$.

Then $\left(I + vu^T\right)\left(I + \lambda vu^T\right) = I + \lambda vu^T + vu^T + \lambda\left(vu^T\right)^2 = I + (1 + \lambda + 3\lambda)\, vu^T$.

Since $vu^T = \begin{pmatrix} 2 & 2 & 2 \\ -1 & -1 & -1 \\ 2 & 2 & 2 \end{pmatrix} \neq O$, the product equals $I$ exactly when $1 + 4\lambda = 0$, i.e. $\lambda = -\dfrac{1}{4}$.

A right inverse of a square matrix is its inverse, so $\left(M^T\right)^{-1} = I - \dfrac{1}{4} vu^T$ and $\lambda = -0.25$.

**Final Answer: -0.25**

## Question 10

**English**

Let $A = \begin{pmatrix} 1 & 1 \\ 0 & 5 \end{pmatrix}$. For each positive integer $n$, let $s_n$ denote the sum of all the entries of $A^n$ and let $t_n$ denote the trace of $A^n$. Then the value of $\lim\limits_{n\to\infty} \dfrac{t_n}{s_n}$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

**हिन्दी**

माना कि $A = \begin{pmatrix} 1 & 1 \\ 0 & 5 \end{pmatrix}$ है। प्रत्येक धनात्मक पूर्णांक $n$ के लिए माना $s_n$, $A^n$ की सभी प्रविष्टियों का योग है तथा $t_n$, $A^n$ का अनुरेख (trace) है। तब $\lim\limits_{n\to\infty} \dfrac{t_n}{s_n}$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

**Solution**

Compute the first powers: $A^2 = \begin{pmatrix} 1 & 1+5 \\ 0 & 25 \end{pmatrix} = \begin{pmatrix} 1 & 6 \\ 0 & 25 \end{pmatrix}$ and $A^3 = \begin{pmatrix} 1 & 1+30 \\ 0 & 125 \end{pmatrix} = \begin{pmatrix} 1 & 31 \\ 0 & 125 \end{pmatrix}$.

The $(1,2)$ entries $1, 6, 31$ are $1$, $1+5$, $1+5+25$, i.e. $\dfrac{5^n-1}{4}$. Claim: $A^n = \begin{pmatrix} 1 & \frac{5^n-1}{4} \\ 0 & 5^n \end{pmatrix}$ for all $n \geq 1$.

The claim holds for $n = 1$. If it holds for $n$, then $A^{n+1} = A^n A = \begin{pmatrix} 1\cdot 1 & 1\cdot 1 + \frac{5^n-1}{4}\cdot 5 \\ 0 & 5^n \cdot 5 \end{pmatrix}$, and $1 + \dfrac{5^{n+1}-5}{4} = \dfrac{5^{n+1}-1}{4}$. So it holds for $n+1$, and by induction for all $n$.

Then $t_n = 1 + 5^n$ and $s_n = 1 + \dfrac{5^n-1}{4} + 5^n = \dfrac{4+5^n-1+4\cdot 5^n}{4} = \dfrac{5^{n+1}+3}{4}$.

$\dfrac{t_n}{s_n} = \dfrac{4\,(5^n+1)}{5^{n+1}+3} = \dfrac{4\,(1+5^{-n})}{5+3\cdot 5^{-n}}$.

As $n \to \infty$, $5^{-n} \to 0$, so the limit is $\dfrac{4}{5} = 0.80$.

**Final Answer: 0.80**

## Question 11

**English**

Let $A = \begin{pmatrix} -1 & 1 & 0 \\ -2 & 2 & 1 \\ 2 & -1 & 2 \end{pmatrix}$. Then the sum of all the entries of $A^{10}$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

**हिन्दी**

माना कि $A = \begin{pmatrix} -1 & 1 & 0 \\ -2 & 2 & 1 \\ 2 & -1 & 2 \end{pmatrix}$ है। तब $A^{10}$ की सभी प्रविष्टियों का योग _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

**Solution**

Let $N = A - I = \begin{pmatrix} -2 & 1 & 0 \\ -2 & 1 & 1 \\ 2 & -1 & 1 \end{pmatrix}$.

$N^2$: row 1 $(-2, 1, 0)$ gives $(4-2,\ -2+1,\ 0+1) = (2, -1, 1)$; row 2 $(-2, 1, 1)$ gives $(4-2+2,\ -2+1-1,\ 0+1+1) = (4, -2, 2)$; row 3 $(2, -1, 1)$ gives $(-4+2+2,\ 2-1-1,\ 0-1+1) = (0, 0, 0)$.

So $N^2 = \begin{pmatrix} 2 & -1 & 1 \\ 4 & -2 & 2 \\ 0 & 0 & 0 \end{pmatrix} \neq O$.

$N^3 = N^2 N$: row 1 $(2, -1, 1)$ gives $(-4+2+2,\ 2-1-1,\ 0-1+1) = (0, 0, 0)$; row 2 is twice row 1, so it also gives 0; row 3 is 0. Hence $N^3 = O$ and $N^k = O$ for $k \geq 3$.

Since $IN = NI$, the binomial theorem applies: $A^{10} = (I+N)^{10} = \sum_{k=0}^{10} \binom{10}{k} N^k = I + 10N + 45N^2$.

The sum of entries is linear: $\text{sum}(I) = 3$, $\text{sum}(N) = -2+1+0-2+1+1+2-1+1 = 1$, $\text{sum}\big(N^2\big) = 2-1+1+4-2+2 = 6$.

Hence the sum of the entries of $A^{10}$ is $3 + 10 \cdot 1 + 45 \cdot 6 = 3 + 10 + 270 = 283$.

**Final Answer: 283**

## Question 12

**English**

Let $A = \begin{pmatrix} 1 & -\sqrt{3} \\ \sqrt{3} & 1 \end{pmatrix}$ and let $A^{2026} = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$. Then the value of $\frac{c}{a}$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

**हिन्दी**

माना कि $A = \begin{pmatrix} 1 & -\sqrt{3} \\ \sqrt{3} & 1 \end{pmatrix}$ है तथा $A^{2026} = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ है। तब $\frac{c}{a}$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

**Solution**

Since $1^2 + \left(\sqrt{3}\right)^2 = 4$, write $A = 2\begin{pmatrix} \frac{1}{2} & -\frac{\sqrt{3}}{2} \\ \frac{\sqrt{3}}{2} & \frac{1}{2} \end{pmatrix} = 2R_{\pi/3}$, where $R_\theta = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}$.

The map $x+iy \mapsto \begin{pmatrix} x & -y \\ y & x \end{pmatrix}$ respects products, so $R_\theta$ corresponds to $\cos\theta + i\sin\theta$ and $A$ to $1+i\sqrt{3}$.

By De Moivre's theorem, $(\cos\theta + i\sin\theta)^n = \cos n\theta + i\sin n\theta$, so $R_\theta^n = R_{n\theta}$ and $A^n = 2^n R_{n\pi/3}$.

For $n = 2026$: $2026 = 3 \cdot 675 + 1$, so $\dfrac{2026\pi}{3} = 675\pi + \dfrac{\pi}{3}$. As 675 is odd, $\cos\left(675\pi + \frac{\pi}{3}\right) = -\cos\frac{\pi}{3} = -\frac{1}{2}$ and $\sin\left(675\pi + \frac{\pi}{3}\right) = -\sin\frac{\pi}{3} = -\frac{\sqrt{3}}{2}$.

So $a = 2^{2026}\left(-\frac{1}{2}\right) = -2^{2025}$ and $c = 2^{2026}\left(-\frac{\sqrt{3}}{2}\right) = -2^{2025}\sqrt{3}$.

Hence $\dfrac{c}{a} = \sqrt{3} = 1.7320\ldots$, which is 1.73 correct to two decimal places.

**Final Answer: 1.73**

## Question 13

**English**

Let $A = \begin{pmatrix} \frac{1}{4} & 1 \\ \frac{1}{16} & -\frac{1}{4} \end{pmatrix}$ and let $I$ be the $2 \times 2$ identity matrix. For each positive integer $n$, let $S_n = I + A + A^2 + \cdots + A^n$. Then the value of $\lim_{n\to\infty}$ (sum of all the entries of $S_n$) is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

**हिन्दी**

माना कि $A = \begin{pmatrix} \frac{1}{4} & 1 \\ \frac{1}{16} & -\frac{1}{4} \end{pmatrix}$ है तथा $I$, $2 \times 2$ तत्समक आव्यूह है। प्रत्येक धनात्मक पूर्णांक $n$ के लिए माना $S_n = I + A + A^2 + \cdots + A^n$ है। तब $\lim_{n\to\infty}$ ($S_n$ की सभी प्रविष्टियों का योग) का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

**Solution**

Compute $A^2$: the $(1,1)$ entry is $\frac{1}{16} + 1 \cdot \frac{1}{16} = \frac{1}{8}$, the $(1,2)$ entry is $\frac{1}{4} \cdot 1 + 1 \cdot \left(-\frac{1}{4}\right) = 0$, the $(2,1)$ entry is $\frac{1}{16} \cdot \frac{1}{4} + \left(-\frac{1}{4}\right) \frac{1}{16} = 0$, and the $(2,2)$ entry is $\frac{1}{16} + \frac{1}{16} = \frac{1}{8}$. So $A^2 = \frac{1}{8}I$.

Hence $A^{2k} = \left(A^2\right)^k = 8^{-k}I$ and $A^{2k+1} = 8^{-k}A$ for $k \geq 0$.

Grouping the terms in pairs, $S_{2m+1} = \sum_{k=0}^{m} \left(A^{2k} + A^{2k+1}\right) = \left(\sum_{k=0}^{m} 8^{-k}\right)(I + A)$.

The geometric series has ratio $\frac{1}{8}$, so $\sum_{k=0}^{m} 8^{-k} = \dfrac{1 - 8^{-(m+1)}}{1 - \frac{1}{8}} \to \dfrac{8}{7}$.

Also $S_{2m} = S_{2m+1} - A^{2m+1} = S_{2m+1} - 8^{-m}A$, and $8^{-m}A \to O$, so both subsequences have the same limit $\dfrac{8}{7}(I + A)$.

$I + A = \begin{pmatrix} \frac{5}{4} & 1 \\ \frac{1}{16} & \frac{3}{4} \end{pmatrix}$, whose entries add up to $\frac{5}{4} + 1 + \frac{1}{16} + \frac{3}{4} = \frac{49}{16}$.

Since the sum of entries is linear, the required limit is $\dfrac{8}{7} \cdot \dfrac{49}{16} = \dfrac{7}{2} = 3.50$.

**Final Answer: 3.50**

## Question 14

**English**

Let $a$ and $b$ be real numbers, and let $A = \begin{pmatrix} 1 & a & b \\ 0 & 1 & a \\ 0 & 0 & 1 \end{pmatrix}$ be such that $A^{10} = \begin{pmatrix} 1 & 2 & 0 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \end{pmatrix}$. Then the value of $b$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

**हिन्दी**

माना कि $a$ तथा $b$ वास्तविक संख्याएँ हैं, तथा $A = \begin{pmatrix} 1 & a & b \\ 0 & 1 & a \\ 0 & 0 & 1 \end{pmatrix}$ इस प्रकार है कि $A^{10} = \begin{pmatrix} 1 & 2 & 0 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \end{pmatrix}$ है। तब $b$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

**Solution**

Pattern: $A^2 = \begin{pmatrix} 1 & 2a & 2b + a^2 \\ 0 & 1 & 2a \\ 0 & 0 & 1 \end{pmatrix}$ and $A^3 = \begin{pmatrix} 1 & 3a & 3b + 3a^2 \\ 0 & 1 & 3a \\ 0 & 0 & 1 \end{pmatrix}$ suggest $A^n = \begin{pmatrix} 1 & na & nb + \frac{n(n-1)}{2}a^2 \\ 0 & 1 & na \\ 0 & 0 & 1 \end{pmatrix}$.

Induction step: multiplying this by $A$, the $(1,3)$ entry becomes $b + na \cdot a + nb + \frac{n(n-1)}{2}a^2 = (n+1)b + \frac{(n+1)n}{2}a^2$, and the $(1,2)$, $(2,3)$ entries become $(n+1)a$. So the formula holds for all $n$.

The same formula follows from $A = I + N$ with $N = \begin{pmatrix} 0 & a & b \\ 0 & 0 & a \\ 0 & 0 & 0 \end{pmatrix}$: $N^2$ has only the $(1,3)$ entry $a^2$, $N^3 = O$, and $I$ commutes with $N$, so $(I+N)^n = I + nN + \binom{n}{2}N^2$.

For $n = 10$ the formula gives $A^{10} = \begin{pmatrix} 1 & 10a & 10b + 45a^2 \\ 0 & 1 & 10a \\ 0 & 0 & 1 \end{pmatrix}$.

Comparing the $(1,2)$ and $(2,3)$ entries with the given matrix: $10a = 2$, so $a = \dfrac{1}{5}$.

Comparing the $(1,3)$ entries: $10b + 45 \cdot \dfrac{1}{25} = 0$, so $10b = -\dfrac{9}{5}$ and $b = -\dfrac{9}{50} = -0.18$.

(Ignoring the $\binom{10}{2}a^2$ term would wrongly give $b = 0$.)

**Final Answer: -0.18**

## Question 15

**English**

Let $A = \begin{pmatrix} 2 & 3 \\ 1 & 0 \end{pmatrix}$ and let $I$ be the $2 \times 2$ identity matrix. If $\alpha$ and $\beta$ are real numbers such that $A^4 + 3A^{-1} = \alpha A + \beta I$, then the value of $\alpha + \beta$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

**हिन्दी**

माना कि $A = \begin{pmatrix} 2 & 3 \\ 1 & 0 \end{pmatrix}$ है तथा $I$, $2 \times 2$ तत्समक आव्यूह है। यदि $\alpha$ तथा $\beta$ ऐसी वास्तविक संख्याएँ हैं कि $A^4 + 3A^{-1} = \alpha A + \beta I$ है, तो $\alpha + \beta$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

**Solution**

$\mathrm{tr}A = 2$ and $\det A = 0 - 3 = -3$, so the characteristic equation is $\lambda^2 - 2\lambda - 3 = 0$ (eigenvalues 3 and $-1$).

By the Cayley-Hamilton theorem, $A^2 - 2A - 3I = O$, i.e. $A^2 = 2A + 3I$.

Then $A^3 = A \cdot A^2 = 2A^2 + 3A = 2(2A + 3I) + 3A = 7A + 6I$.

And $A^4 = A \cdot A^3 = 7A^2 + 6A = 7(2A + 3I) + 6A = 20A + 21I$.

From $A^2 - 2A = 3I$: $A(A - 2I) = 3I$, so $A$ is invertible and $A^{-1} = \dfrac{1}{3}(A - 2I)$; hence $3A^{-1} = A - 2I$.

Adding: $A^4 + 3A^{-1} = (20A + 21I) + (A - 2I) = 21A + 19I$.

Since $A$ is not a multiple of $I$, the matrices $A$ and $I$ are linearly independent, so $\alpha = 21$ and $\beta = 19$ are unique.

Check with eigenvalues: $3^4 + \frac{3}{3} = 82 = 3\alpha + \beta$ and $(-1)^4 - 3 = -2 = -\alpha + \beta$ give the same $\alpha, \beta$.

Hence $\alpha + \beta = 40$.

**Final Answer: 40**

## Question 16

**English**

Let $A$ be a $2 \times 2$ matrix with real entries such that $\mathrm{tr}(A) = 0$ and $\det(A) = -1$, where $\mathrm{tr}(A)$ denotes the trace of $A$. Let $I$ be the $2 \times 2$ identity matrix and $B = A + 2I$. If $p$ and $q$ are real numbers such that $B^6 - 3B^{-1} = pI + qA$, then the value of $p$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

**हिन्दी**

माना कि $A$ वास्तविक प्रविष्टियों वाला एक $2 \times 2$ आव्यूह है जिसके लिए $\mathrm{tr}(A) = 0$ तथा $\det(A) = -1$ है, जहाँ $\mathrm{tr}(A)$, $A$ का अनुरेख (trace) है। माना $I$, $2 \times 2$ तत्समक आव्यूह है तथा $B = A + 2I$ है। यदि $p$ तथा $q$ ऐसी वास्तविक संख्याएँ हैं कि $B^6 - 3B^{-1} = pI + qA$ है, तो $p$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

**Solution**

For a $2 \times 2$ matrix the characteristic equation is $\lambda^2 - (\mathrm{tr}A)\lambda + \det A = 0$, here $\lambda^2 - 1 = 0$.

By the Cayley-Hamilton theorem, $A^2 - I = O$, so $A^2 = I$: $A$ is involutory, and $A^k = I$ for even $k$, $A^k = A$ for odd $k$.

Since $A$ and $2I$ commute, the binomial theorem gives $B^6 = (2I + A)^6 = \sum_{k=0}^{6} \binom{6}{k} 2^{6-k} A^k$.

The even-$k$ terms give $\left(\sum_{k \text{ even}} \binom{6}{k} 2^{6-k}\right) I = \dfrac{(2+1)^6 + (2-1)^6}{2} I = \dfrac{729+1}{2} I = 365I$.

The odd-$k$ terms give $\dfrac{(2+1)^6 - (2-1)^6}{2} A = 364A$. So $B^6 = 365I + 364A$.

Inverse: $(A+2I)(2I-A) = 4I - A^2 = 3I$, so $B$ is invertible and $B^{-1} = \dfrac{1}{3}(2I - A)$; thus $3B^{-1} = 2I - A$.

Hence $B^6 - 3B^{-1} = (365 - 2)I + (364 + 1)A = 363I + 365A$.

$A$ is not a scalar matrix: if $A = cI$ then $\mathrm{tr}A = 2c = 0$ forces $c = 0$ and $\det A = 0 \neq -1$. So $I$ and $A$ are linearly independent, and $p = 363$, $q = 365$ are unique.

(The answer does not depend on which such $A$ is taken; for example $A = \begin{pmatrix} 1 & 2 \\ 0 & -1 \end{pmatrix}$ gives the same $p$ and $q$.)

**Final Answer: 363**

## Question 17

**English**

Let $A$ be a $3 \times 3$ matrix with real entries such that $\det(A - I) = \det(A + I) = \det(A - 2I) = 0$, where $I$ is the $3 \times 3$ identity matrix. Then the value of $\det\left(A^2 - 2A^{-1}\right)$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

**हिन्दी**

माना कि $A$ वास्तविक प्रविष्टियों वाला एक $3 \times 3$ आव्यूह है जिसके लिए $\det(A-I) = \det(A+I) = \det(A-2I) = 0$ है, जहाँ $I$, $3 \times 3$ तत्समक आव्यूह है। तब $\det\left(A^2 - 2A^{-1}\right)$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

**Solution**

$\det(A - \lambda I)$ is a cubic polynomial in $\lambda$ with leading term $-\lambda^3$. It vanishes at $\lambda = 1, -1, 2$, so $\det(A - \lambda I) = -(\lambda - 1)(\lambda + 1)(\lambda - 2)$.

Hence the eigenvalues of $A$ are $1, -1, 2$, the characteristic equation is $\lambda^3 - 2\lambda^2 - \lambda + 2 = 0$, and $\det A = 1 \cdot (-1) \cdot 2 = -2 \neq 0$; so $A$ is invertible.

By the Cayley-Hamilton theorem, $A^3 - 2A^2 - A + 2I = O$, i.e. $A\left(A^2 - 2A - I\right) = -2I$.

Multiplying by $A^{-1}$: $A^{-1} = -\dfrac{1}{2}\left(A^2 - 2A - I\right)$, so $-2A^{-1} = A^2 - 2A - I$.

Therefore $A^2 - 2A^{-1} = A^2 + A^2 - 2A - I = 2A^2 - 2A - I$.

If $AX = \lambda X$ with $X \neq 0$, then $\left(2A^2 - 2A - I\right)X = \left(2\lambda^2 - 2\lambda - 1\right)X$. So the eigenvalues of $2A^2 - 2A - I$ are $2\lambda^2 - 2\lambda - 1$ at $\lambda = 1, -1, 2$, namely $-1,\ 3,\ 3$ (three eigenvalues of a $3 \times 3$ matrix, one for each independent eigenvector of $A$).

The determinant is the product of the eigenvalues: $(-1)(3)(3) = -9$.

(Direct check with eigenvalues of $A^{-1}$: $\lambda^2 - \dfrac{2}{\lambda}$ gives $1 - 2 = -1$, $1 + 2 = 3$, $4 - 1 = 3$, the same values.)

**Final Answer: -9**

## Question 18

**English**

Let $A = \begin{pmatrix} \frac{2}{5} & \frac{4}{5} \\ \frac{3}{10} & \frac{3}{5} \end{pmatrix}$ and let $I$ be the $2 \times 2$ identity matrix. For each positive integer $n$, let $\left(I - \frac{1}{2}A\right)^n = \begin{pmatrix} a_n & b_n \\ c_n & d_n \end{pmatrix}$. Then the value of $\lim\limits_{n\to\infty} a_n$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

**हिन्दी**

माना कि $A = \begin{pmatrix} \frac{2}{5} & \frac{4}{5} \\ \frac{3}{10} & \frac{3}{5} \end{pmatrix}$ है तथा $I$, $2 \times 2$ तत्समक आव्यूह है। प्रत्येक धनात्मक पूर्णांक $n$ के लिए माना $\left(I - \frac{1}{2}A\right)^n = \begin{pmatrix} a_n & b_n \\ c_n & d_n \end{pmatrix}$ है। तब $\lim\limits_{n\to\infty} a_n$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

**Solution**

Compute $A^2$: $(1,1)$ entry $\frac{4}{25} + \frac{4}{5} \cdot \frac{3}{10} = \frac{4}{25} + \frac{6}{25} = \frac{2}{5}$; $(1,2)$ entry $\frac{8}{25} + \frac{12}{25} = \frac{4}{5}$; $(2,1)$ entry $\frac{6}{50} + \frac{9}{50} = \frac{3}{10}$; $(2,2)$ entry $\frac{6}{25} + \frac{9}{25} = \frac{3}{5}$. So $A^2 = A$.

Hence $A^k = A$ for every $k \geq 1$ ($A$ is idempotent).

Since $I$ and $A$ commute, the binomial theorem gives $\left(I - \frac{1}{2}A\right)^n = I + \sum_{k=1}^{n} \binom{n}{k} \left(-\frac{1}{2}\right)^k A^k = I + \left[\sum_{k=1}^{n} \binom{n}{k} \left(-\frac{1}{2}\right)^k\right] A$.

The bracket equals $\left(1 - \frac{1}{2}\right)^n - 1 = 2^{-n} - 1$. So $\left(I - \frac{1}{2}A\right)^n = I + (2^{-n} - 1)\, A$.

Reading the $(1,1)$ entry: $a_n = 1 + (2^{-n} - 1) \cdot \dfrac{2}{5} = \dfrac{3}{5} + \dfrac{2}{5} \cdot 2^{-n}$.

As $n \to \infty$, $2^{-n} \to 0$, so $\lim\limits_{n\to\infty} a_n = \dfrac{3}{5} = 0.60$.

**Final Answer: 0.60**

## Question 19

**English**

Let $B$ and $C$ be $2 \times 2$ real orthogonal matrices, each with determinant 1, whose first columns are $\begin{pmatrix} \frac{4}{5} \\ \frac{3}{5} \end{pmatrix}$ and $\begin{pmatrix} -\frac{12}{13} \\ \frac{5}{13} \end{pmatrix}$, respectively. If $B^T C = \begin{pmatrix} p & q \\ r & s \end{pmatrix}$, where $B^T$ is the transpose of $B$, then the value of $65(p+q)$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

**हिन्दी**

माना कि $B$ तथा $C$, $2 \times 2$ वास्तविक लांबिक (orthogonal) आव्यूह हैं, प्रत्येक का सारणिक 1 है, तथा इनके प्रथम स्तम्भ क्रमशः $\begin{pmatrix} \frac{4}{5} \\ \frac{3}{5} \end{pmatrix}$ तथा $\begin{pmatrix} -\frac{12}{13} \\ \frac{5}{13} \end{pmatrix}$ हैं। यदि $B^T C = \begin{pmatrix} p & q \\ r & s \end{pmatrix}$ है, जहाँ $B^T$, $B$ का परिवर्त है, तो $65(p+q)$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

**Solution**

A real $2 \times 2$ matrix is orthogonal exactly when its columns are perpendicular unit vectors.

If the first column is $(\cos\theta, \sin\theta)^T$, the second column is a unit vector perpendicular to it, so it is $\pm(-\sin\theta, \cos\theta)^T$; the determinant condition $\det = 1$ selects the $+$ sign. So the matrix is $R_\theta = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}$.

Thus $B = R_\alpha$ with $\cos\alpha = \frac{4}{5}$, $\sin\alpha = \frac{3}{5}$, and $C = R_\beta$ with $\cos\beta = -\frac{12}{13}$, $\sin\beta = \frac{5}{13}$ (both first columns are unit vectors).

$B$ is orthogonal, so $B^T = B^{-1} = R_{-\alpha} = \begin{pmatrix} \cos\alpha & \sin\alpha \\ -\sin\alpha & \cos\alpha \end{pmatrix}$.

Multiplying, $B^T C = \begin{pmatrix} \cos\alpha\cos\beta + \sin\alpha\sin\beta & -\cos\alpha\sin\beta + \sin\alpha\cos\beta \\ \cdots & \cdots \end{pmatrix} = \begin{pmatrix} \cos(\beta-\alpha) & -\sin(\beta-\alpha) \\ \sin(\beta-\alpha) & \cos(\beta-\alpha) \end{pmatrix}$ by the compound-angle formulas.

$\cos(\beta-\alpha) = \cos\beta\cos\alpha + \sin\beta\sin\alpha = \left(-\frac{12}{13}\right)\frac{4}{5} + \frac{5}{13}\cdot\frac{3}{5} = \frac{-48+15}{65} = -\frac{33}{65}$.

$\sin(\beta-\alpha) = \sin\beta\cos\alpha - \cos\beta\sin\alpha = \frac{5}{13}\cdot\frac{4}{5} + \frac{12}{13}\cdot\frac{3}{5} = \frac{20+36}{65} = \frac{56}{65}$.

So $p = -\frac{33}{65}$, $q = -\frac{56}{65}$ and $65(p+q) = -33 - 56 = -89$.

**Final Answer: -89**

## Question 20

**English**

Let $A$ and $B$ be $3\times3$ matrices with real entries such that $\text{tr}\left(A^2\right) = 2$, $\text{tr}\left(B^2\right) = 4$ and $\text{tr}\left((A+2B)^2\right) = 4$, where $\text{tr}(X)$ denotes the trace of a square matrix $X$. Then the value of $\text{tr}\left((2A-B)(A+B)\right)$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

**हिन्दी**

माना कि $A$ तथा $B$ वास्तविक प्रविष्टियों वाले $3 \times 3$ आव्यूह हैं जिनके लिए $\text{tr}\left(A^2\right) = 2$, $\text{tr}\left(B^2\right) = 4$ तथा $\text{tr}\left((A+2B)^2\right) = 4$ है, जहाँ $\text{tr}(X)$ वर्ग आव्यूह $X$ का अनुरेख (trace) है। तब $\text{tr}\left((2A-B)(A+B)\right)$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

**Solution**

Matrix multiplication is not commutative, so $(A+2B)^2 = (A+2B)(A+2B) = A^2+2AB+2BA+4B^2$, keeping $AB$ and $BA$ separate.

The trace is linear, and $\text{tr}(XY) = \text{tr}(YX)$ for square matrices: $\text{tr}(XY) = \sum_{i,j} x_{ij}y_{ji} = \text{tr}(YX)$. So $\text{tr}(BA) = \text{tr}(AB)$; call it $t$.

Taking traces: $\text{tr}\left((A+2B)^2\right) = \text{tr}\left(A^2\right) + 4t + 4\,\text{tr}\left(B^2\right)$, i.e. $4 = 2 + 4t + 16$, so $t = -\dfrac{14}{4} = -\dfrac{7}{2}$.

Expand the required product: $(2A-B)(A+B) = 2A^2 + 2AB - BA - B^2$.

Taking traces: $2\cdot 2 + 2t - t - 4 = t = -\dfrac{7}{2}$.

Such matrices exist, e.g. $A = \text{diag}(1,1,0)$ and $B = \begin{pmatrix} -\frac{5}{4} & \frac{1}{4} & 0 \\ -\frac{1}{4} & -\frac{5}{4} & 0 \\ 0 & 0 & 0 \end{pmatrix}$, so the data are consistent.

Hence the value is $-3.50$.

**Final Answer: -3.50**

## Question 21

**English**

Let $A$ be a $3 \times 3$ matrix with real entries such that $\operatorname{tr}\left(A^2\right) = 50$ and $\operatorname{tr}\left(AA^T\right) = 250$, where $\operatorname{tr}(X)$ denotes the trace of a square matrix $X$ and $A^T$ is the transpose of $A$. Then the sum of the squares of all the entries of $A - A^T$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

**हिन्दी**

माना कि $A$ वास्तविक प्रविष्टियों वाला एक $3 \times 3$ आव्यूह है जिसके लिए $\operatorname{tr}\left(A^2\right) = 50$ तथा $\operatorname{tr}\left(AA^T\right) = 250$ है, जहाँ $\operatorname{tr}(X)$ वर्ग आव्यूह $X$ का अनुरेख है तथा $A^T$, $A$ का परिवर्त है। तब $A - A^T$ की सभी प्रविष्टियों के वर्गों का योग _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

**Solution**

Every square matrix splits as $A = S + K$, where $S = \frac{1}{2}\left(A + A^T\right)$ is symmetric and $K = \frac{1}{2}\left(A - A^T\right)$ is skew-symmetric; then $A^T = S - K$ and $A - A^T = 2K$.

For any square $X$, the sum of the squares of its entries is $\operatorname{tr}\left(XX^T\right)$. For $X = 2K$: $\operatorname{tr}\left(2K \cdot 2K^T\right) = -4\operatorname{tr}\left(K^2\right)$ since $K^T = -K$.

Cross terms: $\operatorname{tr}(SK) = \operatorname{tr}\left((SK)^T\right) = \operatorname{tr}\left(K^T S^T\right) = \operatorname{tr}(-KS) = -\operatorname{tr}(KS) = -\operatorname{tr}(SK)$, using $\operatorname{tr}(X^T) = \operatorname{tr}(X)$ and $\operatorname{tr}(KS) = \operatorname{tr}(SK)$. Hence $\operatorname{tr}(SK) = \operatorname{tr}(KS) = 0$.

$A^2 = (S + K)^2 = S^2 + SK + KS + K^2$, so $\operatorname{tr}\left(A^2\right) = \operatorname{tr}\left(S^2\right) + \operatorname{tr}\left(K^2\right) = 50$.

$AA^T = (S + K)(S - K) = S^2 - SK + KS - K^2$, so $\operatorname{tr}\left(AA^T\right) = \operatorname{tr}\left(S^2\right) - \operatorname{tr}\left(K^2\right) = 250$.

Subtracting: $2\operatorname{tr}\left(K^2\right) = -200$, so $\operatorname{tr}\left(K^2\right) = -100$ (and $\operatorname{tr}\left(S^2\right) = 150$).

Therefore the sum of the squares of the entries of $A - A^T$ is $-4(-100) = 400$.

(For example $A = \begin{pmatrix} 5 & 5 & 5 \\ -5 & 5 & 0 \\ -5 & 0 & 10 \end{pmatrix}$ has $\operatorname{tr}\left(A^2\right) = 50$, $\operatorname{tr}\left(AA^T\right) = 250$ and gives 400.)

**Final Answer: 400**

## Question 22

**English**

Let $R = \begin{pmatrix} \cos\frac{2\pi}{9} & -\sin\frac{2\pi}{9} \\ \sin\frac{2\pi}{9} & \cos\frac{2\pi}{9} \end{pmatrix}$ be the matrix of the anticlockwise rotation of the $xy$-plane about the origin through the angle $\dfrac{2\pi}{9}$. For $k = 1, 2, \ldots, 9$, let $(x_k, y_k)$ be the point whose position vector is $R^k \begin{pmatrix} \frac{1}{2} \\ \frac{1}{2} \end{pmatrix}$. Then the value of $\dfrac{1}{9}\displaystyle\sum_{k=1}^{9} x_k^2$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

**हिन्दी**

माना कि $R = \begin{pmatrix} \cos\frac{2\pi}{9} & -\sin\frac{2\pi}{9} \\ \sin\frac{2\pi}{9} & \cos\frac{2\pi}{9} \end{pmatrix}$, $xy$-तल के मूलबिन्दु के परितः कोण $\dfrac{2\pi}{9}$ से वामावर्त घूर्णन का आव्यूह है। $k = 1, 2, \ldots, 9$ के लिए माना $(x_k, y_k)$ वह बिन्दु है जिसका स्थिति सदिश $R^k \begin{pmatrix} \frac{1}{2} \\ \frac{1}{2} \end{pmatrix}$ है। तब $\dfrac{1}{9}\displaystyle\sum_{k=1}^{9} x_k^2$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

**Solution**

Identify the point $(x, y)$ with $z = x + iy$. Then $R\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} x\cos\theta - y\sin\theta \\ x\sin\theta + y\cos\theta \end{pmatrix}$ with $\theta = \frac{2\pi}{9}$ corresponds to $\omega z$, where $\omega = \cos\theta + i\sin\theta$.

Applying $R$ $k$ times multiplies by $\omega^k$, and by De Moivre's theorem $\omega^k = \cos k\theta + i\sin k\theta$; so $R^k$ is the rotation through $k\theta$.

Hence $z_k = x_k + iy_k = \omega^k z_0$ with $z_0 = \frac{1}{2} + \frac{1}{2}i$, and $|z_k|^2 = |z_0|^2 = \frac{1}{4} + \frac{1}{4} = \frac{1}{2}$.

Since $x_k = \dfrac{z_k + \bar{z}_k}{2}$: $x_k^2 = \dfrac{z_k^2 + \bar{z}_k^2 + 2z_k\bar{z}_k}{4} = \dfrac{z_k^2 + \bar{z}_k^2 + 1}{4}$.

$\sum_{k=1}^{9} z_k^2 = z_0^2 \sum_{k=1}^{9} \left(\omega^2\right)^k$. Here $\omega^2 = \cos\frac{4\pi}{9} + i\sin\frac{4\pi}{9} \neq 1$ and $\left(\omega^2\right)^9 = \cos 4\pi + i\sin 4\pi = 1$, so the geometric sum is $\omega^2\dfrac{\omega^{18} - 1}{\omega^2 - 1} = 0$.

Taking conjugates, $\sum \bar{z}_k^2 = 0$ as well.

So $\sum_{k=1}^{9} x_k^2 = \dfrac{0 + 0 + 9}{4} = \dfrac{9}{4}$ and the mean is $\dfrac{1}{9} \cdot \dfrac{9}{4} = \dfrac{1}{4} = 0.25$.

(In words: over a full set of nine equally spaced rotations the mean of $x^2$ is half the squared distance from the origin.)

**Final Answer: 0.25**

## Question 23

**English**

For real numbers $k$ and $\beta$, consider the system of linear equations

$$x - y + 2z = 1, \qquad 2x + y + kz = 2, \qquad x + 3y - z = \beta$$

in the unknowns $x, y, z$. If the system has infinitely many solutions, then the value of $k + \beta$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

**हिन्दी**

वास्तविक संख्याओं $k$ तथा $\beta$ के लिए अज्ञात $x, y, z$ में रैखिक समीकरण निकाय

$$x - y + 2z = 1, \qquad 2x + y + kz = 2, \qquad x + 3y - z = \beta$$

पर विचार कीजिए। यदि निकाय के अनन्त हल हैं, तो $k + \beta$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

**Solution**

Write the system as $AX = C$ with $A = \begin{pmatrix} 1 & -1 & 2 \\ 2 & 1 & k \\ 1 & 3 & -1 \end{pmatrix}$, $C = \begin{pmatrix} 1 \\ 2 \\ \beta \end{pmatrix}$.

If $X_1 \neq X_2$ are both solutions, then $A(X_1 - X_2) = O$ with $X_1 - X_2 \neq O$: the homogeneous system has a non-trivial solution, which happens only when $\det A = 0$.

Expanding along the first row: $\det A = 1\,(1 \cdot (-1) - 3k) - (-1)\,(2 \cdot (-1) - k) + 2\,(6 - 1) = -1 - 3k - 2 - k + 10 = 7 - 4k$.

So $k = \dfrac{7}{4}$. Then rows 1 and 2 of $A$ are independent, so $A$ has rank 2 and its third row is a combination of the first two.

Solve $(1, 3, -1) = a(1, -1, 2) + b\left(2, 1, \frac{7}{4}\right)$: from the first two entries $a + 2b = 1$ and $-a + b = 3$, so $3b = 4$, $b = \frac{4}{3}$, $a = -\frac{5}{3}$; the third entry checks: $2a + \frac{7}{4}b = -\frac{10}{3} + \frac{7}{3} = -1$.

So equation 3 minus $a$ times equation 1 minus $b$ times equation 2 has left side 0; its right side $\beta - a - 2b$ must also be 0, otherwise the system is inconsistent.

$\beta = a + 2b = -\frac{5}{3} + \frac{8}{3} = 1$. With $k = \frac{7}{4}$, $\beta = 1$ the system reduces to two independent equations in three unknowns, which have infinitely many solutions.

Hence $k + \beta = \dfrac{7}{4} + 1 = \dfrac{11}{4} = 2.75$.

**Final Answer: 2.75**

## Question 24

**English**

Let $S$ denote the set of all $3 \times 3$ symmetric matrices $A$, each of whose entries belongs to the set $\{-1, 0, 1\}$, such that $\operatorname{tr}\left(A^2\right) = 4$, where $\operatorname{tr}(X)$ denotes the trace of a square matrix $X$. Then the number of elements of $S$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

**हिन्दी**

माना कि $S$ उन सभी $3 \times 3$ सममित आव्यूहों $A$ का समुच्चय है, जिनकी प्रत्येक प्रविष्टि समुच्चय $\{-1, 0, 1\}$ से है, तथा $\operatorname{tr}\left(A^2\right) = 4$ है, जहाँ $\operatorname{tr}(X)$ वर्ग आव्यूह $X$ का अनुरेख है। तब $S$ के अवयवों की संख्या _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

**Solution**

For a symmetric matrix $A^T = A$, so $A^2 = AA^T$ and $\operatorname{tr}\left(A^2\right) = \operatorname{tr}\left(AA^T\right) = \sum_{i,j} a_{ij}^2$, the sum of the squares of all nine entries.

Each off-diagonal value appears twice ($a_{ij} = a_{ji}$), so $\operatorname{tr}\left(A^2\right) = \sum_i a_{ii}^2 + 2\sum_{i<j} a_{ij}^2$.

A symmetric matrix is fixed by its 3 diagonal entries and the 3 entries above the diagonal. With entries in $\{-1, 0, 1\}$, each square is 0 or 1. Let $d$ be the number of non-zero diagonal entries ($0 \le d \le 3$) and $p$ the number of non-zero entries above the diagonal ($0 \le p \le 3$). The condition is

$d + 2p = 4$.

Solutions: $(d, p) = (2, 1)$ or $(0, 2)$; $(4, 0)$ is impossible since $d \leq 3$.

Case $(2, 1)$: choose the two non-zero diagonal places in $\binom{3}{2} = 3$ ways and their signs in $2^2 = 4$ ways; choose the non-zero upper entry in 3 ways and its sign in 2 ways (its mirror entry is then fixed). Count $= 12 \times 6 = 72$.

Case $(0, 2)$: diagonal all zero; choose two of the three upper places in 3 ways and their signs in 4 ways. Count $= 12$.

Total: $72 + 12 = 84$.

**Final Answer: 84**

## Question 25

**English**

Let $N_1$ denote the number of $3 \times 3$ matrices, each of whose entries is 0 or 1, having exactly four entries equal to 1, and let $N_2$ denote the number of those matrices that are symmetric. Then the value of $\dfrac{N_1}{N_2}$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

**हिन्दी**

माना कि $N_1$ उन $3 \times 3$ आव्यूहों की संख्या है, जिनकी प्रत्येक प्रविष्टि $0$ या $1$ है तथा जिनकी ठीक चार प्रविष्टियाँ $1$ के बराबर हैं, और $N_2$ इनमें से सममित आव्यूहों की संख्या है। तब $\dfrac{N_1}{N_2}$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

**Solution**

A 0-1 matrix with exactly four 1's is fixed by the set of positions of its 1's, a selection of 4 of the 9 positions. So $N_1 = \binom{9}{4} = \dfrac{9 \cdot 8 \cdot 7 \cdot 6}{24} = 126$.

For a symmetric matrix, the positions split into the 3 diagonal places and 3 off-diagonal pairs $\{(i, j), (j, i)\}$, $i < j$; the two entries of a pair are equal.

If $d$ diagonal entries and $p$ pairs are 1, the number of 1's is $d + 2p = 4$ with $0 \leq d \leq 3$, $0 \leq p \leq 3$. So $(d, p) = (2, 1)$ or $(0, 2)$ ($d = 4$ is impossible).

Case $(2, 1)$: select 2 of the 3 diagonal places and 1 of the 3 pairs: $\binom{3}{2}\binom{3}{1} = 9$.

Case $(0, 2)$: select 2 of the 3 pairs: $\binom{3}{2} = 3$.

So $N_2 = 9 + 3 = 12$ and $\dfrac{N_1}{N_2} = \dfrac{126}{12} = \dfrac{21}{2} = 10.50$.

**Final Answer: 10.50**

## Question 26

**English**

A $3 \times 3$ matrix $A$ is transformed into the $3 \times 3$ identity matrix $I$ by the following elementary row operations, applied in the given order:

(i) $R_1 \leftrightarrow R_2$; (ii) $R_1 \to R_1 - 2R_3$; (iii) $R_3 \to 2R_3$; (iv) $R_2 \to R_2 + 2R_1$; (v) $R_3 \to R_3 + 2R_2$.

Then the sum of all the entries of $A$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

**हिन्दी**

एक $3 \times 3$ आव्यूह $A$ को निम्नलिखित प्रारम्भिक पंक्ति संक्रियाएँ, दिए गए क्रम में, लगाने पर $3 \times 3$ तत्समक आव्यूह $I$ प्राप्त होता है:

(i) $R_1 \leftrightarrow R_2$; (ii) $R_1 \to R_1 - 2R_3$; (iii) $R_3 \to 2R_3$; (iv) $R_2 \to R_2 + 2R_1$; (v) $R_3 \to R_3 + 2R_2$.

तब $A$ की सभी प्रविष्टियों का योग _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

**Solution**

Each elementary row operation equals left multiplication by the matrix $E$ obtained by doing that operation on $I$. Let $E_1, \ldots, E_5$ correspond to (i)-(v). Then $E_5E_4E_3E_2E_1A = I$.

Hence $A$ is invertible with $A^{-1} = E_5E_4E_3E_2E_1$ (the same operations applied to $I$ give $A^{-1} = \begin{pmatrix} 0 & 1 & -2 \\ 1 & 2 & -4 \\ 2 & 4 & -6 \end{pmatrix}$), and $A = \left(A^{-1}\right)^{-1} = E_1^{-1}E_2^{-1}E_3^{-1}E_4^{-1}E_5^{-1}$.

The inverse of each elementary operation is elementary: (i) is undone by $R_1 \leftrightarrow R_2$, (ii) by $R_1 \to R_1 + 2R_3$, (iii) by $R_3 \to \frac{1}{2}R_3$, (iv) by $R_2 \to R_2 - 2R_1$, (v) by $R_3 \to R_3 - 2R_2$.

$A = E_1^{-1}E_2^{-1}E_3^{-1}E_4^{-1}E_5^{-1}I$ means: starting from $I$, apply $E_5^{-1}$ first and $E_1^{-1}$ last.

$E_5^{-1}$ $(R_3 \to R_3 - 2R_2)$: $\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & -2 & 1 \end{pmatrix}$; then $E_4^{-1}$ $(R_2 \to R_2 - 2R_1)$: $\begin{pmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ 0 & -2 & 1 \end{pmatrix}$.

Then $E_3^{-1}$ $(R_3 \to \frac{1}{2}R_3)$: $\begin{pmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ 0 & -1 & \frac{1}{2} \end{pmatrix}$; then $E_2^{-1}$ $(R_1 \to R_1 + 2R_3)$: $\begin{pmatrix} 1 & -2 & 1 \\ -2 & 1 & 0 \\ 0 & -1 & \frac{1}{2} \end{pmatrix}$.

Finally $E_1^{-1}$ $(R_1 \leftrightarrow R_2)$: $A = \begin{pmatrix} -2 & 1 & 0 \\ 1 & -2 & 1 \\ 0 & -1 & \frac{1}{2} \end{pmatrix}$.

Check: applying (i)-(v) to this $A$ gives $\begin{pmatrix} 1 & -2 & 1 \\ -2 & 1 & 0 \\ 0 & -1 & \frac{1}{2} \end{pmatrix} \to \begin{pmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ 0 & -1 & \frac{1}{2} \end{pmatrix} \to \begin{pmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ 0 & -2 & 1 \end{pmatrix} \to \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & -2 & 1 \end{pmatrix} \to I$.

Sum of entries of $A$: $(-2 + 1 + 0) + (1 - 2 + 1) + \left(0 - 1 + \frac{1}{2}\right) = -1 + 0 - \frac{1}{2} = -1.50$.

**Final Answer: -1.50**

## Question 27

**English**

Let $M$ be the $4 \times 4$ matrix whose $(i, j)$ entry is $\min\{i, j\}$ for $i, j = 1, 2, 3, 4$. Then the sum of all those entries of $M^{-1}$ that do not lie on the principal diagonal is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

**हिन्दी**

माना कि $M$ वह $4 \times 4$ आव्यूह है जिसकी $(i, j)$ प्रविष्टि $i, j = 1, 2, 3, 4$ के लिए $\min\{i, j\}$ है। तब $M^{-1}$ की उन सभी प्रविष्टियों का योग, जो मुख्य विकर्ण पर नहीं हैं, _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

**Solution**

Here $M = \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & 2 & 2 & 2 \\ 1 & 2 & 3 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix}$.

Apply $R_4 \to R_4 - R_3$, then $R_3 \to R_3 - R_2$, then $R_2 \to R_2 - R_1$ (each uses the original rows, so do them in this order). These do not change the determinant and give $U = \begin{pmatrix} 1 & 1 & 1 & 1 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 \end{pmatrix}$. Hence $\det M = \det U = 1 \neq 0$, so $M$ is invertible.

Equivalently, row $i$ of $M$ is the sum of the first $i$ rows of $U$, i.e. $M = LU$ with $L$ the lower triangular matrix of ones; and $L = U^T$. So $M = U^T U$ and $\det M = \det U^T \det U = 1$.

$U^{-1} = \begin{pmatrix} 1 & -1 & 0 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & 1 & -1 \\ 0 & 0 & 0 & 1 \end{pmatrix}$: indeed $U \cdot U^{-1} = I$, because each row of $U$ is $(0, \ldots, 0, 1, \ldots, 1)$ and the columns of $U^{-1}$ take differences of neighbouring entries.

Then $M^{-1} = U^{-1} \left(U^T\right)^{-1} = U^{-1} \left(U^{-1}\right)^T$.

Multiplying: $M^{-1} = \begin{pmatrix} 2 & -1 & 0 & 0 \\ -1 & 2 & -1 & 0 \\ 0 & -1 & 2 & -1 \\ 0 & 0 & -1 & 1 \end{pmatrix}$. Check one product: row 2 of $M$, $(1, 2, 2, 2)$, times column 2 of $M^{-1}$, $(-1, 2, -1, 0)$, is $-1 + 4 - 2 = 1$, and times column 1, $(2, -1, 0, 0)$, is $2 - 2 = 0$.

The off-diagonal entries are six entries equal to $-1$ and six zeros, so their sum is $-6$.

**Final Answer: -6**

## Question 28

**English**

Let $A = \begin{pmatrix} 3 & -1 & 0 \\ -1 & 1 & -1 \\ 0 & -1 & 2 \end{pmatrix}$. Then the sum of all the entries of $A^{-1}$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

**हिन्दी**

माना कि $A = \begin{pmatrix} 3 & -1 & 0 \\ -1 & 1 & -1 \\ 0 & -1 & 2 \end{pmatrix}$ है। तब $A^{-1}$ की सभी प्रविष्टियों का योग _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

**Solution**

Start from the augmented matrix $[A \,|\, I] = \left(\begin{array}{ccc|ccc} 3 & -1 & 0 & 1 & 0 & 0 \\ -1 & 1 & -1 & 0 & 1 & 0 \\ 0 & -1 & 2 & 0 & 0 & 1 \end{array}\right)$.

$R_1 \leftrightarrow R_2$ and then $R_1 \to -R_1$: $\left(\begin{array}{ccc|ccc} 1 & -1 & 1 & 0 & -1 & 0 \\ 3 & -1 & 0 & 1 & 0 & 0 \\ 0 & -1 & 2 & 0 & 0 & 1 \end{array}\right)$.

$R_2 \to R_2 - 3R_1$: the second row becomes $(0, 2, -3 \,|\, 1, 3, 0)$.

$R_3 \to 2R_3$: $(0, -2, 4 \,|\, 0, 0, 2)$; then $R_3 \to R_3 + R_2$: $(0, 0, 1 \,|\, 1, 3, 2)$.

$R_2 \to R_2 + 3R_3$: $(0, 2, 0 \,|\, 4, 12, 6)$; then $R_2 \to \frac{1}{2}R_2$: $(0, 1, 0 \,|\, 2, 6, 3)$.

$R_1 \to R_1 + R_2 - R_3$: $(1, -1+1-0, 1+0-1 \,|\, 0+2-1, -1+6-3, 0+3-2) = (1, 0, 0 \,|\, 1, 2, 1)$.

The left block is now $I$, so $A^{-1} = \begin{pmatrix} 1 & 2 & 1 \\ 2 & 6 & 3 \\ 1 & 3 & 2 \end{pmatrix}$.

Check: row 1 of $A$ times the columns of $A^{-1}$ gives $(3-2,\ 6-6,\ 3-3) = (1, 0, 0)$; similarly for the other rows.

Sum of all entries: $(1+2+1) + (2+6+3) + (1+3+2) = 4 + 11 + 6 = 21$.

**Final Answer: 21**

## Question 29

**English**

For real numbers $a$ and $b$, consider the planes

$$P_1 :\ x + 2y - z = 2, \qquad P_2 :\ 2x - y + az = 1, \qquad P_3 :\ 3x + y + z = b.$$

Suppose that the system of the three equations of $P_1$, $P_2$, $P_3$ in the unknowns $x, y, z$ has infinitely many solutions, so that the three planes have a common line $L$. If $(\alpha, \beta, \gamma)$ is the foot of the perpendicular drawn from the origin to $L$, then the value of $\alpha\beta$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

**हिन्दी**

वास्तविक संख्याओं $a$ तथा $b$ के लिए तलों

$$P_1 :\ x + 2y - z = 2, \qquad P_2 :\ 2x - y + az = 1, \qquad P_3 :\ 3x + y + z = b$$

पर विचार कीजिए। माना कि अज्ञात $x, y, z$ में $P_1$, $P_2$, $P_3$ के तीनों समीकरणों से बने निकाय के अनन्त हल हैं, जिससे तीनों तलों की एक उभयनिष्ठ रेखा $L$ है। यदि मूलबिन्दु से $L$ पर डाले गए लम्ब का पाद $(\alpha, \beta, \gamma)$ है, तो $\alpha\beta$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

**Solution**

Let $D = \det \begin{pmatrix} 1 & 2 & -1 \\ 2 & -1 & a \\ 3 & 1 & 1 \end{pmatrix} = 1(-1-a) - 2(2-3a) - 1(2+3) = 5a - 10.$

If $D \neq 0$ the system has a unique solution, so infinitely many solutions require $D = 0$, i.e. $a = 2$.

With $a = 2$ the left sides satisfy $P_3 = P_1 + P_2$: $(1+2,\ 2-1,\ -1+2) = (3, 1, 1)$. Subtracting the first two equations from the third leaves $0 = b - 3$, so the system is consistent only if $b = 3$; then the third equation is redundant.

The normals $(1, 2, -1)$ and $(2, -1, 2)$ of $P_1$, $P_2$ are not parallel, so $P_1 \cap P_2$ is a line $L$, and it lies in $P_3$; this gives the infinitely many solutions.

Direction of $L$: $\vec{n}_1 \times \vec{n}_2 = (2 \cdot 2 - (-1)(-1),\ (-1) \cdot 2 - 1 \cdot 2,\ 1 \cdot (-1) - 2 \cdot 2) = (3, -4, -5)$.

A point of $L$: put $z = 0$; $x + 2y = 2$ and $2x - y = 1$ give $x = \frac{4}{5}$, $y = \frac{3}{5}$. So $Q = \left(\frac{4}{5}, \frac{3}{5}, 0\right)$ and $L:\ \vec{r} = \left(\frac{4}{5}, \frac{3}{5}, 0\right) + t(3, -4, -5)$.

The foot $F$ of the perpendicular from $O$ satisfies $\overrightarrow{OF} \cdot (3, -4, -5) = 0$: $\left(\frac{4}{5} + 3t\right) 3 + \left(\frac{3}{5} - 4t\right)(-4) + (-5t)(-5) = \frac{12}{5} - \frac{12}{5} + 50t = 0$, so $t = 0$.

Hence $F = Q = \left(\frac{4}{5}, \frac{3}{5}, 0\right)$, i.e. $\alpha = \frac{4}{5}$, $\beta = \frac{3}{5}$, $\gamma = 0$, and the distance from $O$ to $L$ is 1.

$$\alpha\beta = \frac{12}{25} = 0.48.$$

**Final Answer: 0.48**